

STREAMING LIFECYCLE ANALYTICS DASHBOARD

Introduction

In the modern music industry, streaming platforms have transformed how songs gain popularity and sustain their presence among listeners. Playlists, especially curated charts such as regional Top 50 lists, play a critical role in shaping listener exposure and artist success. However, understanding how songs behave over time within these playlists remains a complex challenge.

This study focuses on the Spanish Top 50 playlist and aims to analyse the lifecycle of songs from entry to exit. The objective is to understand how long songs remain relevant, how quickly they reach peak performance, and how various content attributes influence their longevity. By examining lifecycle patterns, churn behaviour, and content characteristics, this work provides insights into the dynamics of playlist-driven music consumption.

Dataset Description

The dataset used in this analysis consists of daily snapshots of Spain’s Top 50 playlist, containing a total of 27,800 records. Each record represents a song’s position on a given day along with its associated metadata.

The dataset includes features such as song title, artist name, popularity score, duration, album type, number of tracks in the album, and whether the content is explicit. Temporal information is captured through date and ranking position, enabling a time-series analysis of song performance.

The data is complete and well-structured, allowing for reliable lifecycle construction and statistical analysis. Preprocessing steps ensured consistent formatting of dates, normalization of song identifiers, and creation of unique song-level tracking.

date	position	song	artist	popularity	duration_ms	album_type	total_tracks	is_explicit
#####	1	Supernova	Saiko	84	190909	single	1	FALSE
#####	2	EL TONTO	Lola Indigo	83	187986	album	11	FALSE
#####	3	AcrÃ³stico	Shakira	81	170785	single	1	FALSE
#####	4	BESO	ROSALÍA & Rauw Alejandro	96	194543	single	3	FALSE
#####	5	El Merengue	Marshmello & Manuel Turizo	91	189668	single	1	FALSE
#####	6	Mami Chula	Jhayco & Quevedo	85	220600	single	1	FALSE
#####	7	Classy 101	Feid & Young Miko	95	195986	single	1	TRUE

Methodology

The analysis follows a structured pipeline designed to extract meaningful lifecycle patterns from the dataset.

Initially, data preprocessing was performed to clean and standardize the dataset. Dates were converted into appropriate formats, and unique song identifiers were created by combining song and artist information. This allowed consistent tracking of songs across multiple days.

Next, lifecycle metrics were constructed for each song. These included entry date, exit date, total number of days on the playlist, peak position, and time taken to reach that peak. Based on these metrics, songs were further categorized into lifecycle stages such as new entry, growth, peak, mature, and decline.

To understand playlist dynamics, churn analysis was performed by calculating daily entries and exits. This provided insights into how frequently songs are replaced and how stable the playlist remains over time.

Additionally, content-based analysis was conducted to evaluate how attributes such as explicit content, album type, song duration, and album size influence retention and performance.

Finally, all findings were visualized using analytical plots to identify trends, relationships, and anomalies within the dataset.

Results and Analysis

The analysis reveals several important patterns governing playlist behaviour and song performance.

The lifecycle of songs demonstrates a gradual progression rather than instant success. As seen in Figure 1, songs in the peak stage exhibit the highest average popularity, while new entries start at comparatively lower levels and improve over time. This indicates that songs typically build momentum before reaching their maximum impact.



Figure 1

The stability of the playlist is further highlighted by the churn analysis. Figure 2 shows that daily churn remains consistently low, with occasional spikes. These spikes likely correspond to external factors such as major releases or seasonal trends. Overall, the playlist exhibits strong retention with limited daily turnover.

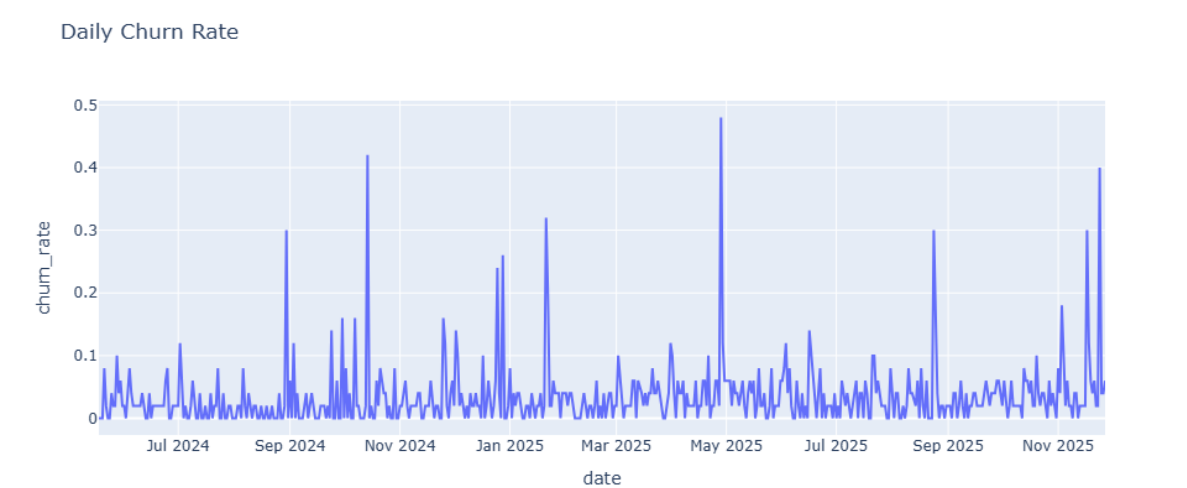


Figure 2

When examining the relationship between song duration and retention (Figure 3), no strong correlation is observed. Songs of varying lengths appear across all retention levels, suggesting that duration alone is not a significant factor in determining playlist longevity.

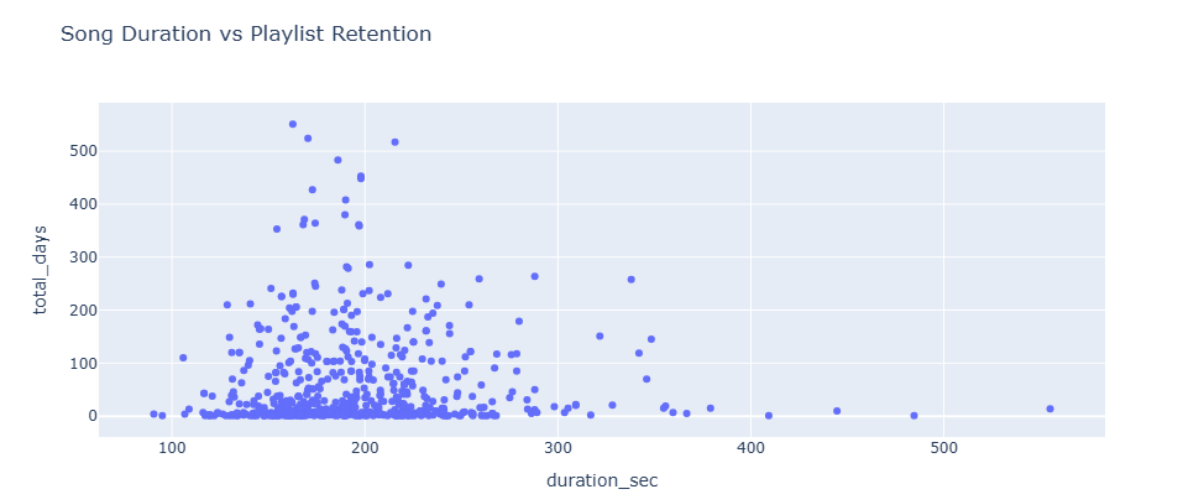


Figure 3

Content type analysis reveals subtle but meaningful trends. Figure 4 compares explicit and non-explicit songs, showing that both categories have similar distributions, with only marginal differences in retention. This indicates that explicit content does not significantly hinder performance.

Explicit vs Non-Explicit Song Longevity

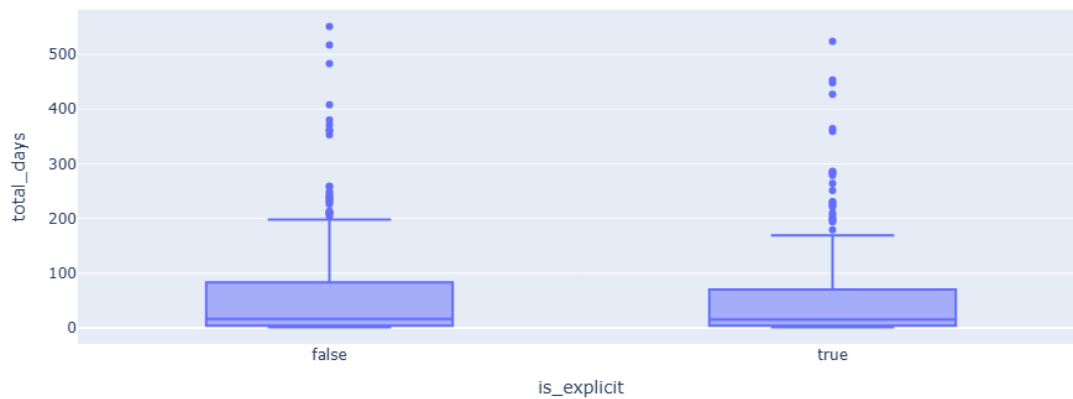


Figure 4

A more pronounced difference is observed when comparing singles and album tracks. As shown in Figure 5, singles tend to have higher median retention and more consistent performance compared to album tracks. This supports the idea that focused releases perform better in playlist environments.

Single vs Album Longevity

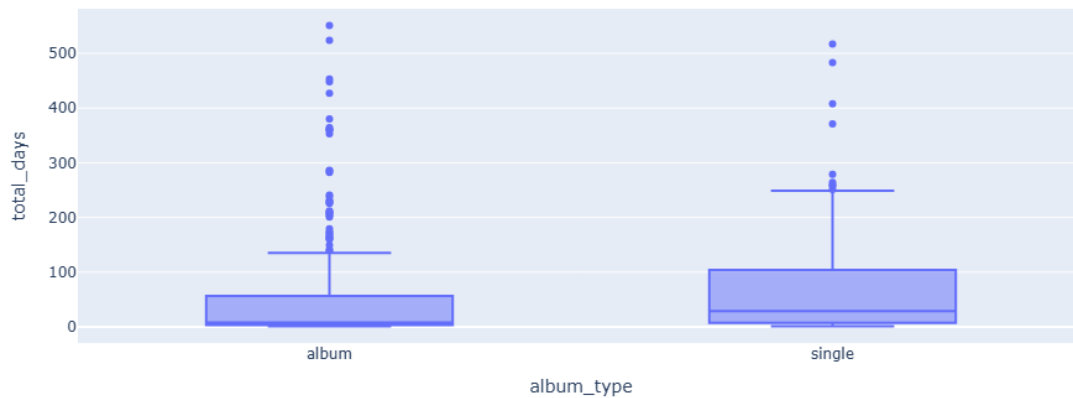


Figure 5

Monthly churn analysis (Figure 6) further confirms the stability of the playlist while highlighting occasional peaks. These peaks suggest periodic disruptions, likely driven by new releases or industry cycles.

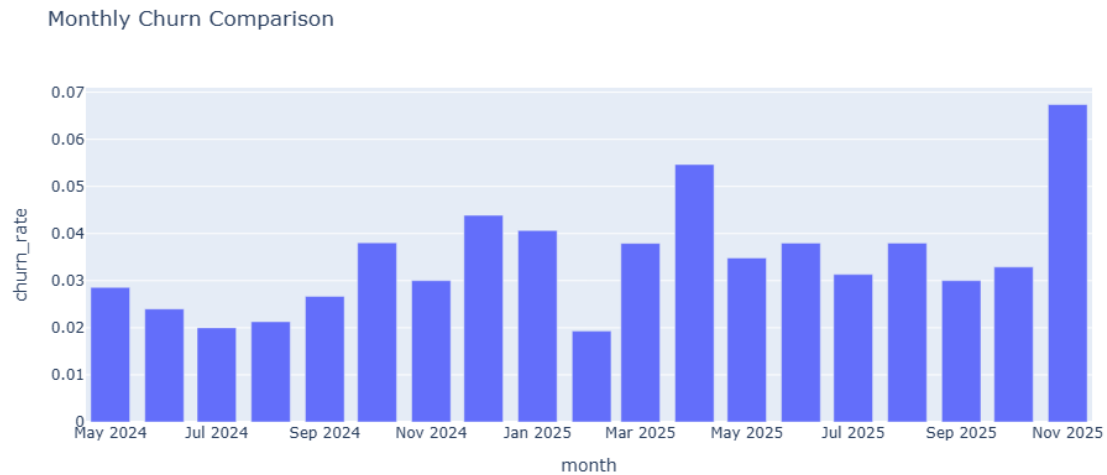


Figure 6

The analysis of popularity around peak timing (Figure 7) shows that songs generally maintain high popularity levels before and after reaching their peak. This reinforces the idea of sustained engagement rather than abrupt rises and falls.

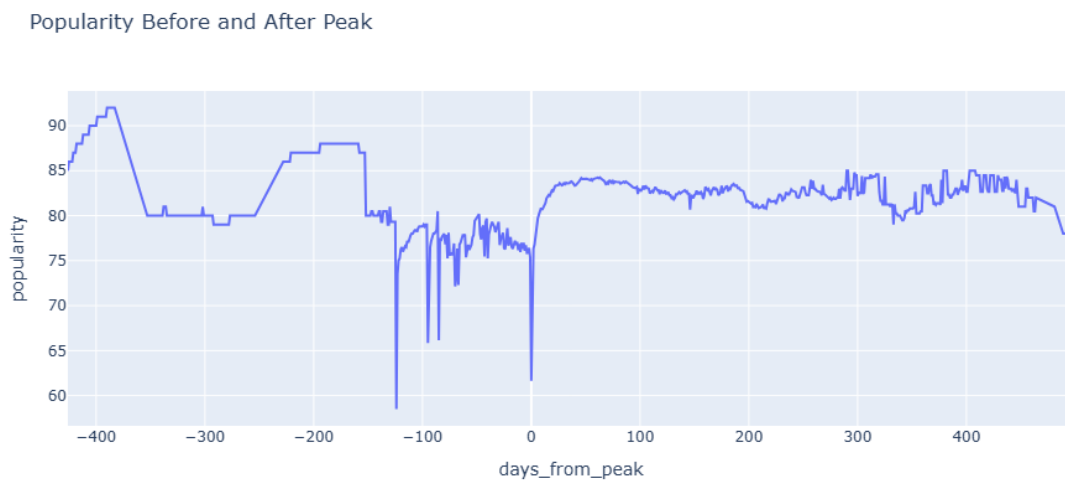


Figure 7

Finally, the relationship between album size and retention (Figure 8) indicates that larger albums do not necessarily lead to better performance. In fact, smaller collections often show more consistent retention, suggesting that listener attention is concentrated on individual tracks rather than entire albums.

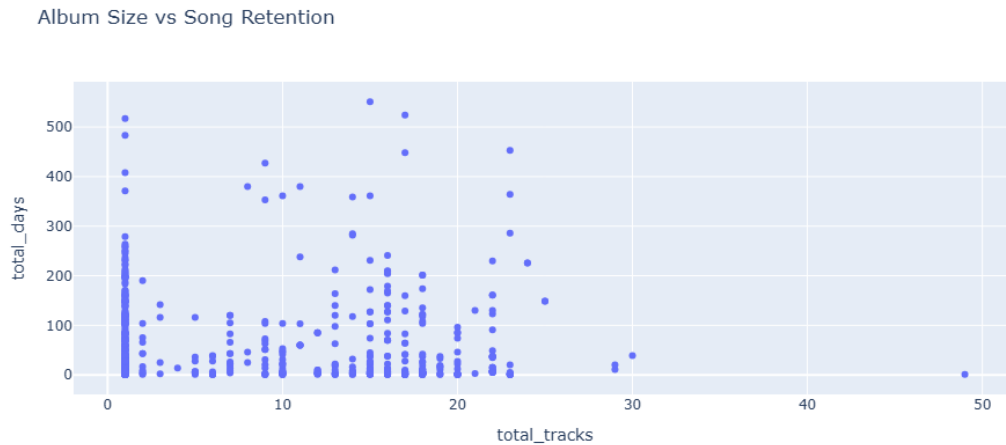


Figure 8

Key Performance Indicators

The key performance indicators derived from the analysis provide a comprehensive view of playlist behaviour and content performance. On average, songs remain on the playlist for 56.42 days, indicating a relatively strong retention and sustained listener engagement. The average time taken for a song to reach its peak position is 9.51 days, suggesting that songs generally build momentum gradually rather than achieving immediate popularity.

The playlist itself exhibits a low churn rate of 3.46%, which reflects a high level of stability, with only a small number of songs being replaced on a daily basis. At the same time, the retention stability index, measured through a standard deviation of 88.30, highlights significant variability in song lifecycles. This indicates that while some tracks dominate the playlist for extended periods, others experience much shorter lifespans, pointing towards a hit-driven ecosystem.

When examining content characteristics, non-explicit songs show an average retention of 60.37 days, while explicit songs demonstrate a very similar performance with an average of 60.88 days. This minimal difference suggests that explicit content does not significantly impact a song's longevity on the playlist.

Furthermore, the analysis reveals that singles tend to outperform album tracks, with a single-to-album longevity ratio of 1.20. This indicates that singles remain on playlists for a longer duration, reinforcing the effectiveness of focused, track-level release strategies over broader album-based approaches.

Average Days on Playlist: 56.42 days

Entry-to-Peak Time: 9.51 days

Playlist Churn Rate: 3.46%

Retention Stability Index (Std Dev): 88.30

Explicit vs Non-Explicit Lifecycle:

Non-Explicit Songs: 60.37 days

Explicit Songs: 60.88 days

Single vs Album Longevity Ratio: 1.20

Discussion

Overall, the Spanish Top 50 playlist demonstrates a stable yet competitive ecosystem. Songs are not immediately replaced, allowing them time to grow and establish popularity. At the same time, only a select few manage to sustain long-term dominance, creating a clear distinction between high-performing hits and short-lived entries.

Content characteristics play a nuanced role in performance. While explicit content does not significantly impact retention, release strategy—particularly the choice between singles and albums—has a noticeable effect. Similarly, album size appears to dilute attention rather than enhance it.

These findings highlight the importance of sustained engagement, strategic release planning, and focus on individual track performance.

Streamlit Dashboard Implementation

To complement the analytical findings, an interactive dashboard was developed using Streamlit. The application provides a user-friendly interface for exploring lifecycle patterns, playlist dynamics, and content behaviour in real time.

The dashboard integrates custom CSS styling to create a modern, visually engaging interface with neon-themed design elements, gradient typography, and glassmorphism-inspired components. Interactive filters allow users to analyse data based on date range, lifecycle stage, content type, and album classification.

Visualizations are powered by Plotly, enabling dynamic exploration of lifecycle curves, churn trends, and content comparisons. Key performance indicators are displayed prominently, providing immediate insights into playlist behaviour.

The application is designed to be both analytical and intuitive, bridging the gap between raw data and actionable insights.

[**Deployment Link**](#)

[**GitHub Repository**](#)



Conclusion

This study provides a comprehensive analysis of playlist lifecycle behaviour within the Spanish music market. The findings demonstrate that the system is characterized by high stability, gradual content growth, and a strong emphasis on sustained engagement.

Songs typically build popularity over time, rather than achieving immediate success, while playlist churn remains low, allowing top-performing tracks to maintain their positions. The market is distinctly hit-driven, with a small number of songs dominating for extended periods.

Strategically, the results highlight the importance of single-track releases, consistent promotion, and focus on individual content performance. These insights can be valuable for artists, record labels, and streaming platforms seeking to optimize their strategies in a competitive digital ecosystem.